Threat-Shape: changes after the review

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The lasso

Previously, when the lasso was released, the tool recomputed a t-SNE on the enclosed incidents alone and drew it in place of the global projection. This was a mistake for several reasons. The subset had been chosen by looking at a t-SNE, so the second projection started from groups that the first one had already separated, and it produced a new representation with no direct relation to the original data. This distorted the view of the data and lost its context.

Now the lasso is only a selection. The enclosed incidents stay highlighted in their original position, the others are dimmed, and the other four views (map, contrast panel, timeline, impact score) update on that selection. This also makes it possible to compare different clusters with each other. We added two controls to the view: a switch between direct t-SNE and PCA + t-SNE, used for the comparison presented in the next section, and a zoom, which makes it easier to examine clusters and the individual points that represent the incidents.

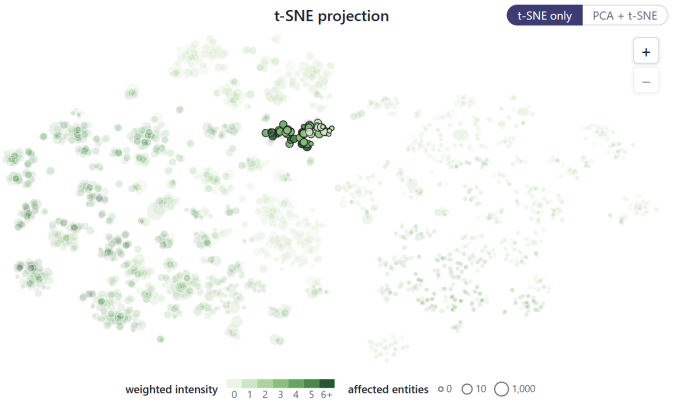


Figure 1. View B after a lasso on the larger group of incidents against Russia: the 65 incidents stay in the global projection, and the rest of the corpus is dimmed.

The contrast panel

In the first version, each row compared the selection made in the other views against the rest of the world (or against a specific country) on indicators chosen among the most representative ones, so the rows changed with every selection. This serious flaw made a coherent and rigorous comparative analysis impossible. There was also a readability problem: each row compared only two values and their difference, but these numbers were not shown explicitly. They could only be read from the scale at the bottom of the chart, through lengths and colours that did not convey the information precisely.

For consistency with the original design, we kept the structure of the chart, but once we understood these flaws we made substantial changes, so that the view is now clearer and can be considered valid. In each row the empty dot represents the comparison group, the filled dot the selection, and the colour shows the sign of the difference (red if the indicator is more frequent in the selection, blue if it is less frequent). To make the information easier to read, each percentage is now written next to its own dot, and the difference in percentage points above the segment that joins them. There are six rows, and they stay the same for every selection. We chose the most important and frequent indicators: who attacks (state-affiliated actor, non-state group), what kind of attack (data theft, disruption) and against whom (state institutions, critical infrastructure). Below 20 incidents all rows are dimmed, with a note warning that the reading is only indicative.

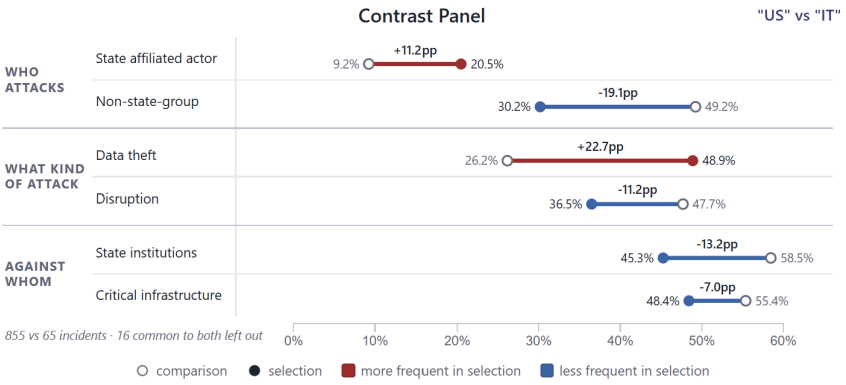


Figure 2. View D: the United States compared with Italy, with the two percentages and their difference for each indicator.

PCA + t-SNE against t-SNE alone

Why PCA was present and what we hadn't checked

We reduced the dimensions using, in sequence, PCA and t-SNE because we found that it is a well-documented practice. Indeed, we can cite several resources:

- Van der Maaten and Hinton reduce each dataset to 30 dimensions to speed up the computation of pairwise distances and to suppress some noise [1]: "In all of our experiments, we start by using PCA to reduce the dimensionality of the data to 30. This speeds up the computation of pairwise distances between the datapoints and suppresses some noise without severely distorting the interpoint distances. We then use each of the dimensionality reduction techniques to convert the 30-dimensional representation to a two-dimensional map and we show the resulting map as a scatterplot".
- The scikit-learn documentation recommends a reduction to about 50 dimensions when the number of features is very large, for the same two reasons as before [2]: "It is highly recommended to use another dimensionality reduction method (e.g. PCA for dense data or TruncatedSVD for sparse data) to reduce the number of dimensions to a reasonable amount (e.g. 50) if the number of features is very high. This will suppress some noise and speed up the computation of pairwise distances between samples".

Our dataset has 127 columns, including 123 binary indicators (the one-hot encoding of 14 categorical variables) and 4 numerical variables, all standardized. This is not a very large number; fitting t-SNE directly on the 127 columns takes about 15 seconds, and our 20-component cutoff (50.7% of the variance) was lower than both suggested values.

We had not yet checked whether the information removed by PCA was actually just noise. For both layouts, we used the same method, TSNE(random_state=42), keeping all the other settings at their default values (perplexity 30 and 1,000 iterations), just as we did in the course exercises. We then compared both layouts with the original 127 standardized columns. For each incident, we considered the 20 closest incidents in the original data as its actual nearest neighbours. Trustworthiness measures how many of the neighbours found are correct, while continuity measures how many of the actual neighbours are preserved. We call direct the layout computed by t-SNE on the 127 columns, and staged the one computed by PCA to 20 components followed by t-SNE.

The issue. Fitting t-SNE on the 127 standardized columns and fitting it on 20 principal components produce two different representations of the same data, because PCA changes the space t-SNE receives. We measured how different the two are, why they differ, and which one is closer to the original data.

What changes the result? PCA first rotates the 127 standardized columns onto new axes, the principal components, and then keeps only the first 20. The rotation alone changes nothing for t-SNE, which depends only on the distances between incidents: with all 127 components the distances are identical to the original ones, and t-SNE reaches a trustworthiness of 0.960, against 0.962 on the original columns. The cut is what changes the input. With 20 components only 54% of each incident's 20 true neighbours are still its neighbours in the space t-SNE receives, so almost half of the neighbourhood is lost before t-SNE starts. Fitted directly on the 127 columns, t-SNE computes similarities on the original indicators, each with the same weight after standardization.

Global fidelity

Layout	Trustw.	Contin.	True neighbours kept
PCA 20 + t-SNE	0.916	0.949	42.8%
t-SNE only	**0.962**	**0.967**	**58.4%**

Components kept	20	40	60	90	127
Variance kept	51%	69%	83%	97%	100%
Trustworthiness	0.916	0.939	0.956	0.960	0.962

- The direct layout is better on both kinds of error, and its trustworthiness is higher at every perplexity tested. The difference is +0.049, +0.050, +0.045, +0.043 and +0.033 for perplexity values of 5, 15, 30, 50 and 100 respectively. Two runs of the same process differ by only 0.0006.
- The problem comes from reducing the number of components too much, not from using PCA itself. When using 90 components, the two methods produce almost the same results.
- 71% of the incidents are better represented in the direct layout, while 16% are better represented in the staged layout. The improvement is even greater with higher severity: +12 percentage points of true neighbours kept for a weighted intensity of 0 to 1, +20 for 4 and above.

What is each map organized by?

For each of the 98 binary indicators present in at least 20 incidents, we checked how many of an incident's 20 nearest neighbours shared the same indicator. We compared the results in the original data with those in both maps.

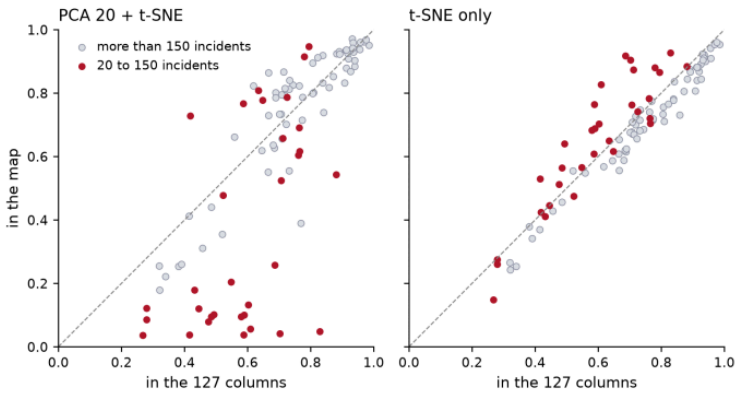


Figure 4. One dot per indicator. On the diagonal the map reproduces the data; below it, the incidents carrying the indicator are scattered.

- The direct layout stays closer to the original data, with an average difference of 0.054 and a correlation of 0.93. The staged layout differs more, with a gap of 0.144 and a correlation of 0.78.
- PCA + t-SNE does a poor job of representing rare indicators. For indicators present in 20–50 incidents, the share of matching neighbours is 0.64 in the original data, drops to 0.26 in the staged map, and reaches 0.75 in the direct map. This happens because rare indicators contribute less to the overall variance, so they are more likely to be removed when PCA reduces the number of components. However, these indicators are important because they help distinguish one type of operation from another.
- At the same time, PCA + t-SNE makes some common features stand out more than they do in the original data. For example, the state response indicator increases from 0.68 to 0.85. This makes the map look clearer and more organized, even though it does not represent all the original information as accurately.

Clusters

Both maps contain around 100 clusters, identified using DBSCAN with the same settings, but fewer than half of these clusters match between the two maps. In the staged layout, 60% of an incident's actual nearest neighbours belong to the same cluster, compared with 75% in the direct layout. In the 15 cases where one map splits a cluster that remains together in the other, the original data always supports that separation. In fact, 96–99% of the actual nearest neighbours belong to the same half of the split, compared with around 50% expected by chance. This means that neither map creates artificial separations: instead, each one reveals differences in the data that the other map groups together.

Case studies



Figure 5. The incidents against Russia in the two layouts, with the two groups as found on the staged one.

Russia. The main finding that the tool was designed to identify appears in both maps. The 220 incidents against Russia are divided into two groups: 37 and 78 incidents in the staged layout, and 36 and 65 in the direct layout. Of these, 34 incidents in the smaller group and 64 in the larger group are shared by both maps. The smaller group mainly consists of denial-of-service attacks lasting just one day, with no data stolen (+72 percentage points more frequent than in the other incidents against Russia). The larger group is mainly made up of intrusions where data was stolen (+79 percentage points more frequent than in the other incidents

against Russia). The main difference between the maps is how far apart these two groups appear. The silhouette score is 0.83 in the staged layout, 0.44 in the direct layout and just 0.08 in the original 127 columns. This means the groups look much more clearly separated in the staged map. However, t-SNE does not preserve the meaning of distances between separate clusters, and the PCA step makes this apparent separation even larger.

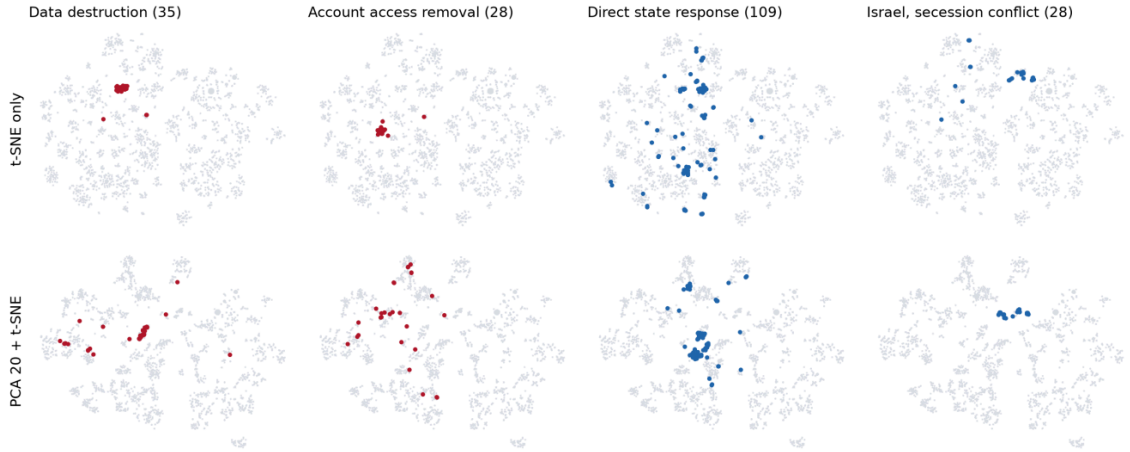


Figure 6. Four groups in the direct layout (top) and in the staged one (bottom). Red: groups only the direct layout draws as a cluster. Blue: groups the staged layout draws more compactly.

Groups that appear only in the direct layout. The 35 incidents involving data destruction, with an average intensity of 4.1 and mostly linked to the Russia-Ukraine conflict, form a single group in the direct layout, with 32 incidents clustered together. In the staged layout, however, they are spread across six different clusters. The same pattern appears for incidents involving account access removal (25 of 28 grouped together in the direct layout, compared with just 3 in the staged one), the use of external remote services to gain access (23 of 25 versus 3), and impacts lasting several months (37 of 58 versus 6).

Where the staged layout performs better. Incidents involving a direct state response, as well as two groups of incidents linked to offline conflicts (28 against Israel and 24 against Ukraine), are more closely grouped in the staged layout. These groups reflect real patterns in the data, but they appear more spread out in the direct layout.

Other countries. When we repeated the Russia analysis for the countries most frequently targeted, we identified 15 groups in the staged layout and 13 in the direct one, with only 7 groups in common. This means that roughly half of the groups identified for each country depend on which layout is used. Russia is the exception, as it is the only case where the two layouts fully agree.

Conclusion. On our data the PCA stage costs more than it gives. t-SNE on the 127 standardized columns is the more faithful picture and brings out small, specific groups of incidents, often severe ones, that the 20-component cut scatters. The staged layout is tidier on the dominant variables, partly beyond what the data support. The projection view therefore opens on the direct layout and keeps the staged one available as a second picture to check a finding against. Fidelity is measured against the 127 columns, the space the direct layout is fitted on. With 90 components, however, the staged pipeline reaches the same score, so the measure penalizes the cut, not PCA as such.

Resources

[1]. L. van der Maaten, G. Hinton. Visualizing Data using t-SNE. Journal of Machine Learning Research 9, 2008, pp. 2579-2605.

[2]. scikit-learn documentation, sklearn.manifold.TSNE . https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html